

$$w_2^{(2)} = 0.18233(0.9) - \frac{0.02}{3} (5439.4160)$$

$$= 0.1641 - 36.2628$$

$$= -36.0987$$

$$w_{\text{Ridge}}^{(2)} = (4.5867, -12.3949, -36.0987)$$

Lasso Regression

Lasso update

$$w_j^{\text{new}} = w_j - \alpha \left[\frac{1}{n} \sum e_i x_{ij} + \lambda \text{sign}(w_j) \right]$$

$$\text{sign}(w_j) = \begin{cases} +1; & w_j > 0 \\ -1; & w_j < 0 \end{cases}$$

Iteration 1

$$w_1 = w_2 = -0.03$$

$$\therefore \text{sign}(w_j) = -1$$

update w_0

No regularization

$$w_0^{(1)} = 5.0016$$

update w_1

$$\begin{aligned}w_1^{(1)} &= -0.03 - 0.02 \left[\frac{-11.65}{3} + 5(-1) \right] \\&= -0.03 - 0.02 [-3.8833 - 5] \\&= -0.03 + 0.17766 \\&= 0.1477\end{aligned}$$

update w_2

$$\begin{aligned}w_2^{(1)} &= -0.03 - 0.02 \left[\frac{-31.40}{3} + 5(-1) \right] \\&= -0.03 - 0.02 [-10.4667 - 5] \\&= -0.03 + 0.30933 \\&= 0.2793\end{aligned}$$

$$[5.0016, 0.1477, 0.2793]$$

Iteration 2

$$w_1^{(1)} = 0.1477 > 0$$

$$w_2^{(1)} = 0.2793 > 0$$

$$\therefore \text{sign}(w_1) = \text{sign}(w_2) = +1$$

$$\hat{y} = 5.0016 + 0.1477x_1 + 0.2793x_2$$

New predictions

$$\hat{y}_1 = 5.0016 + 0.1416(35) + 0.2793(80) = 32.5166$$

$$\hat{y}_2 = 31.0399$$

$$\hat{y}_3 = 37.3649$$

New errors

$$e_1 = 32.5166 - 1.81 = 30.7066$$

$$e_2 = 31.0399 - 1.22 = 29.8199$$

$$e_3 = 37.3649 - 1.71 = 35.6549$$

Compute gradient score

$$\sum e_i = 30.7066 + 29.8199 + 35.6549 = 96.1815$$

$$\sum e_i X_{ii} = 30.7066(35) + 29.8199(25) + 35.6549(30)$$

$$= 2880.8773$$

$$\sum e_i X_{2i} = 30$$

update $w_0(2)$

update

$$\sum e_i X_{2i} = 30.7066(80) + 29.8199(80) \\ + 35.6549(100) \\ = 8409.6160$$

update w_0

$$w_0^{(2)} = 5.0016 - \frac{0.02}{3}(96.1815) \\ = 4.3604$$

update w_1

$$w_1 > 0$$

$$\text{sign}(w_1) = +1$$

$$w_1^{(2)} = 0.14766 - 0.02 \left[\frac{2880.8773}{3} \right]$$

$$= 0.14766 - 0.02 \left[963.2924 \right]$$

$$= -19.2182$$

$$= -19.2182$$

update w_2

$$w_2 > 0$$

$$\text{sign}(w_2) = +1$$

$$\begin{aligned} w_2^{(2)} &= 0.2793 - 0.02 \left[\frac{8407.6160}{3} + 5 \right] \\ &= 0.2793 - 0.02 [2802.5387 + 5] \\ &= -55.8714 \end{aligned}$$

$$[4.3604, -19.2182, -55.8714]$$

Final result

Linear regression (One iteration)

$$[5.0000; -0.0300; -0.0300]$$



$$[5.0018; 0.0477; 0.1793]$$

Ridge regression (2 iterations)

$$[5.0000; -0.0300; -0.0300]$$



$$[5.0016; 0.0507; 0.1823]$$



$$[4.5867; -12.3979; -36.0987]$$

Lasso regression (2 iterations)

$$[5.0000; -0.0300; -0.0300]$$



$$[5.0016; 0.1477; 0.2793]$$



$$[4.3604; -19.2182; -55.8714]$$